

Pythagoras' Theorem & Trigonometry

Extra practice test 2 · 27 marks total

Name: _____ Date: _____ Score: _____ / 27

Section A — Pythagoras' theorem

A1. Point A is 8 km east and 6 km north of point B. Find the direct distance from A to B.

[2 marks]

A2. A ship sails 24 km east, then 7 km south, from its starting point. Use Pythagoras' theorem to find how far the ship now is from its starting point.

[3 marks]

A3. A triangle has sides of length 9 cm, 12 cm and 15 cm. Use Pythagoras' theorem to show that this triangle is right-angled.

[2 marks]

Section B — SOH CAH TOA basics

B1. A student says: "To find a missing angle, I should always use $\sin^{-1}$, no matter which two sides I know." Explain why this is incorrect.

[1 mark]

B2. In a right-angled triangle you know the adjacent side and want to find the opposite side. Which SOH CAH TOA ratio should you use, and write the rearranged formula for the opposite side.

[1 mark]

B3. True or False: $\cos 60^\circ = \sin 30^\circ$. Show a calculation to support your answer.

[1 mark]

Section C — Finding missing sides

C1. From a point on the ground 20 m from the base of a tower, the angle of elevation to the top of the tower is 34° . Find the height of the tower, to 1 decimal place.

[3 marks]

C2. A plane is flying at a height of 1200 m. The angle of depression from the plane to a landmark on the ground is 8° . Find the horizontal distance from the plane to the landmark, to the nearest metre.

[2 marks]

C3. A rope is attached to the top of a flagpole and anchored to the ground, making an angle of 63° with the ground. If the rope is 10 m long, find the height of the flagpole, to 1 decimal place.

[2 marks]

Section D — Finding missing angles

D1. A ramp is 5 m long and rises to a height of 1.2 m. Find the angle the ramp makes with the ground, to 1 decimal place.

[2 marks]

D2. A surveyor stands 45 m from the base of a cliff. The top of the cliff is 62 m above the ground. Find the angle of elevation to the top of the cliff, to 1 decimal place.

[3 marks]

Section E — Applying Pythagoras & trigonometry

E1. An isosceles triangle has two equal sides of length 13 cm and a base of 10 cm. Find the perpendicular height of the triangle by splitting the base into two equal halves.

[2 marks]

E2. A ladder 6 m long leans against a wall, reaching 5.2 m up the wall. (a) Find the distance from the foot of the ladder to the wall. (b) Find the angle the ladder makes with the ground. Give both answers to 1 decimal place.

[3 marks]